EXECUTIVE REPORT

Online Retail Transaction Analysis

Where the value comes from, what leaks away, and what to do next

UK online gift retailer, 1 Dec 2010 to 9 Dec 2011 | 541,909 transaction lines | Prepared by Muneeb Waince, October 2026

The answer in short. Net transaction value was **£9.75M**. Gross transaction value (£10.67M) is 9.4% higher because £896,812 was cancelled. Value builds into autumn, depends on a core group of customers, and 16.5% of it cannot be tied to a customer. Details follow in the executive summary and the action plan.

How to read the counts in this report

Count	Value	How to read it
Transaction lines (rows)	541,909	Each row is one product line on an invoice, not an order and not a customer.
Invoices	25,900	19,960 have gross-value lines; 3,836 are cancellation invoices; 2,104 have neither (zero-price or adjustment lines only).
Cancellation invoices and rows	3,836 invoices, 9,288 rows	3,836 distinct invoice numbers starting with "C" contain 9,288 lines (2.4 lines per invoice). Invoice counts and row counts are never mixed.
Identified customers	4,372	4,338 bought (have gross-value lines). 34 did not: 33 appear only on cancellation invoices and 1 has only a zero-price line.
Lines without CustomerID	135,080 (24.9%)	These lines hold £1,755,277 of gross value (16.5%) that cannot be linked to a customer.

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Evidence labels used throughout: **[Direct]** computed directly from the data; **[Indirect]** follows from documentation or a convention; **[No evidence]** or **[Hypothesis]** not shown by this dataset, so treated as a hypothesis. Amounts are in pounds sterling (per the dataset documentation).

Where to start. Executives: read the executive summary and the action plan (about five minutes). Analysts: Sections 1 to 4 hold the evidence, Section 6 lists assumptions and data handling, and the appendices hold exact definitions (A), verification (B) and technical scope (C).

Executive summary

£9.75M Net transaction value	**£10.67M** Gross transaction value (9.4% above net)	**8.4%** Cancelled value as share of gross (3,836 invoices)
84.6% Share of gross value from the United Kingdom	**4,338** Customers who bought (of 4,372 identified)	**74.6%** Share of identified-customer gross value from the top 20% of customers by value

Bottom line. Over about twelve months the business recorded **£9.75M of net transaction value**. Gross transaction value of £10.67M is £918,937 higher (9.4% above net, or 8.6% of gross) because £896,812 was cancelled and £22,124 was written off as bad debt. **Use net value for decisions.** **[Direct]**

Findings, business impact and recommendations

Finding	Business impact	Recommendation	Evidence
Gross value overstates performance: £918,937 gap. A top-10 product (23843, £168,470 gross) has £0 net.	Rankings and totals built on gross value can mislead decisions.	**R1.** Report net beside gross and adopt the definitions in Appendix A.	**[Direct]**
Value builds into autumn: September to November hold 34.9% of gross value; November is 2.2 times the January to August monthly average.	Stock, staff and cash are stretched in three months. Only one year of data.	**R3.** Plan from August and confirm the pattern with 2012 data.	**[Direct]** pattern **[No evidence]** repeats
A core group carries the value: 23.5% of customers (Champions) hold 64.4% of identified gross value; 212 frequent buyers have lapsed (£393,145).	Losing a few core buyers hurts most; lapsed frequent buyers are a recoverable pool.	**R4.** Key-account cover for top customers and a win-back list.	**[Direct]** size **[No evidence]** win-back works
Cancelled value is concentrated: the top 1% of cancellation invoices (38) hold 64.9% of £896,812. The only 2 lines above 5,000 units (£245,653) were cancelled within 12 to 16 minutes.	Very large orders can inflate gross value. Order value alone is a weak predictor of cancellation.	**R2.** Confirm any line above 5,000 units before dispatch; review invoices above P99 (£4,821).	**[Direct]** facts **[Hypothesis]** rule works
£1,755,277 (16.5%) of gross value has no CustomerID.	This value cannot be tied to buyers, which limits retention analysis.	**R5.** Find out why IDs are missing, then fix capture.	**[Direct]** size **[No evidence]** cause

What this report cannot tell you. Profit (there is no cost data), why orders were cancelled, whether the autumn peak and customer behaviour repeat (one year of 2010-11 data), and how today's market compares (Section 6).

1. Performance over time

Question: When does the business generate its value, and how stable is the trend?
Answer: Gross value runs at about £0.5M to £0.8M a month until August, then climbs to £1.51M in November, 2.2 times the January to August average (£0.69M). September to November hold 34.9% of gross value. **[Direct]**

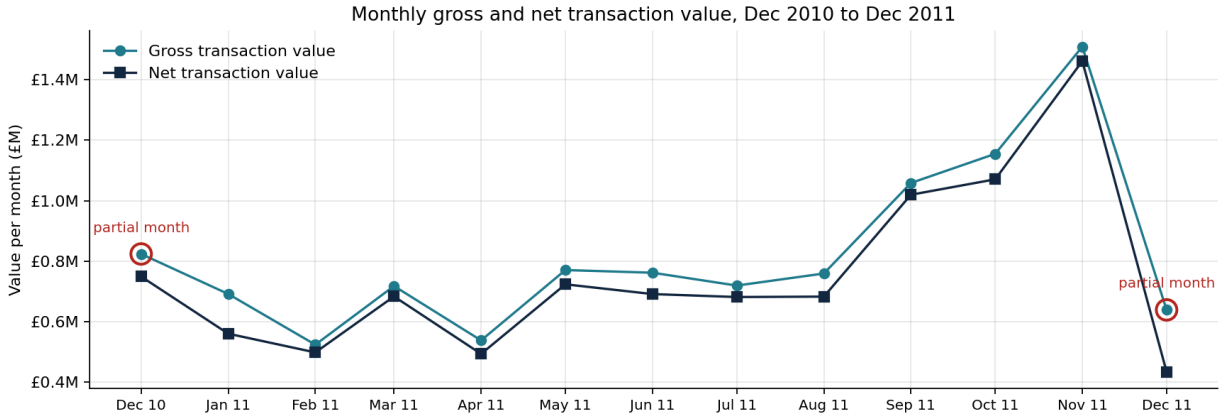


Figure 1. Monthly gross and net transaction value in pounds (£M). Source: UCI Online Retail dataset; author calculations.

- Gross value rose +39.4% in September and peaked at £1.51M in November 2011. **[Direct]**
- Net value is below gross in every month, by 3.2% to 32.1% of gross; the widest gaps are in the two months with the largest cancelled orders.
- The first and last months are partial (no records after 23 Dec 2010 and none after 09 Dec 2011), so their changes are not like-for-like.

Month	Gross value	Net value	Gross change	Net change
Dec 2010*	£823,746.14	£748,957.02		
Jan 2011	£691,364.56	£560,000.26	-16.1%	-25.2%
Feb 2011	£523,631.89	£498,062.65	-24.3%	-11.1%
Mar 2011	£717,639.36	£683,267.08	+37.1%	+37.2%
Apr 2011	£537,808.62	£493,207.12	-25.1%	-27.8%
May 2011	£770,536.02	£723,333.51	+43.3%	+46.7%
Jun 2011	£761,739.90	£691,123.12	-1.1%	-4.5%
Jul 2011	£719,221.19	£681,300.11	-5.6%	-1.4%
Aug 2011	£759,138.38	£682,680.51	+5.6%	+0.2%
Sep 2011	£1,058,590.17	£1,019,687.62	+39.4%	+49.4%
Oct 2011	£1,154,979.30	£1,070,704.67	+9.1%	+5.0%
Nov 2011	£1,509,496.33	£1,461,756.25	+30.7%	+36.5%
Dec 2011*	£638,792.68	£433,668.01	-57.7%	-70.3%

* Partial month. Changes are month on month, computed from unrounded values.

So what. The business is a three-month peak business inside a flat year: September to November produced £3.72M of £10.67M gross value. Whether this repeats each year is **[No evidence]**, because the dataset covers one year only.

2. Where value comes from: countries and products

Question: Which markets and products matter, and do the rankings survive cancellations?
Answer: The United Kingdom provides 84.6% of gross value. Product 22423 (REGENCY CAKESTAND 3 TIER) is first on both gross and net value, but one gross top-10 product falls to £0 net. The ten best products hold only 8.4% of merchandise net value. **[Direct]**

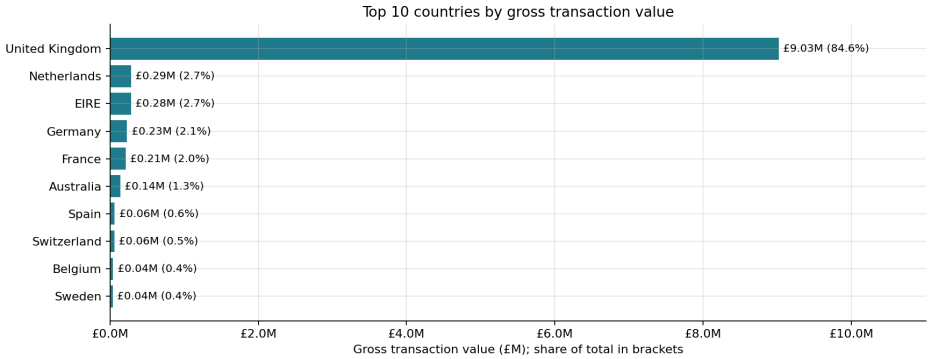


Figure 2. Top 10 countries by gross transaction value (£M), with share of total. Source: UCI Online Retail dataset; author calculations.

- The United Kingdom provides £9,025,222 (84.6%); the next five markets together provide 10.7%. **[Direct]**
- The United Kingdom net value is £8,187,806. Net and gross use different filters, so the two figures must not be compared as one metric.
- Concentration is expected for a UK-registered retailer; whether international value can grow is **[No evidence]**.

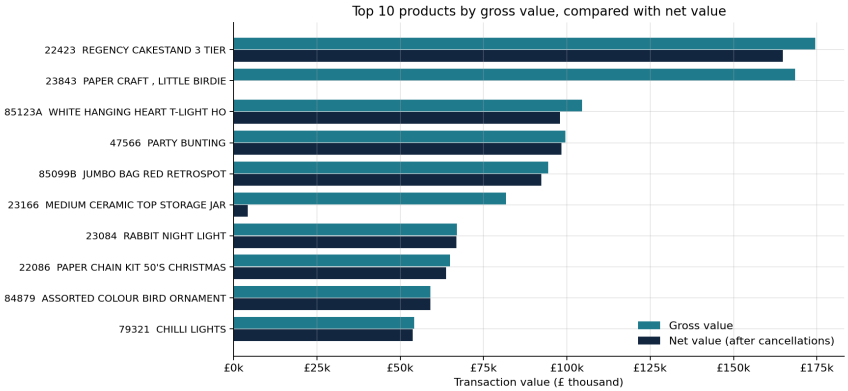


Figure 3. Top 10 products by gross value, with net value after cancellations (£ thousand). Source: UCI Online Retail dataset; author calculations.

- 22423 is first on gross (£174,485) and net (£164,762). **[Direct]**
- 23843 shows £168,470 gross but £0 net: its only order was cancelled. 23166 drops from £81,701 to £4,221.
- The ten best products by net value hold 8.4% of merchandise net value, so the catalogue is broad rather than dependent on a few hits.

Product	Description	Gross value	Net value	Cancelled or returned
22423	REGENCY CAKESTAND 3 TIER	£174,484.74	£164,762.19	£9,722.55
23843	PAPER CRAFT , LITTLE BIRDIE	£168,469.60	£0.00	£168,469.60
85123A	WHITE HANGING HEART T-LIGHT HOLDER	£104,518.80	£97,894.50	£6,624.30
47566	PARTY BUNTING	£99,504.33	£98,302.98	£1,201.35
85099B	JUMBO BAG RED RETROSPOT	£94,340.05	£92,356.03	£1,984.02
23166	MEDIUM CERAMIC TOP STORAGE JAR	£81,700.92	£4,221.28	£77,479.64
23084	RABBIT NIGHT LIGHT	£66,964.99	£66,756.59	£208.40
22086	PAPER CHAIN KIT 50'S CHRISTMAS	£64,952.29	£63,791.94	£1,160.35
84879	ASSORTED COLOUR BIRD ORNAMENT	£59,094.93	£58,959.73	£135.20
79321	CHILLI LIGHTS	£54,117.76	£53,768.06	£349.70

DOT, M and POST are excluded from the product view as likely postage and manual items; this rule needs business confirmation.

3. Customers

Question: Who generates the value, and how concentrated is it?
Answer: A core group does: 23.5% of customers (Champions: recent and frequent buyers) hold 64.4% of identified gross value. Ranked by value alone, the top 20% of customers hold 74.6%. No single customer exceeds 3.1%. **[Direct]**

The analysis covers the 4,338 customers who bought. Of the 4,372 identified customers, 34 have no gross-value line (33 appear only on cancellation invoices; 1 has only a zero-price line). Lines without a CustomerID (£1,755,277, 16.5% of gross value) cannot be attributed to any customer.

Measure	Value	Evidence
Top 20% of customers by value	868 customers = 74.6% of identified gross value	**[Direct]**
Ten largest customers	17.3% of identified gross value (14.4% of all gross value)	**[Direct]**
Largest single customer	3.1% of identified gross value	**[Direct]**
Repeat customers (2 or more invoices)	65.6% of customers	**[Direct]**
Average versus median invoice (gross)	£534.40 versus £303.84: a few large invoices lift the average	**[Direct]**
Wholesale customers	The dataset documentation says many customers are wholesalers; the data has no customer-type field	**[Indirect]**

Two different ways to look at the same customers

Top 20% by value ranks customers by total gross value only, whenever they bought. **Champions** is a behaviour segment: customers whose last order was recent *and* who order frequently. A customer can be in one group, both or neither.

Group	Customers	Gross value	What it means
In both groups	591	£5,196,676	58.1% of Champions and 68.1% of the top 20%: the most valuable core.
Top 20% by value only	277	£1,452,762	High total value but not recent and frequent: large infrequent buyers or lapsing accounts.
Champions only	427	£546,604	Recent and frequent but smaller baskets: reliable, with growth potential.

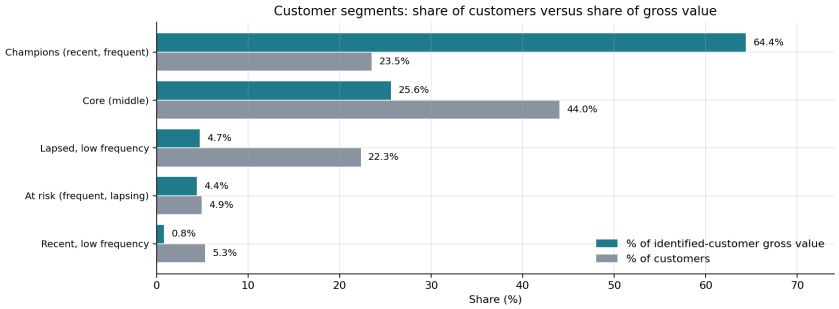


Figure 4. Customer segments: share of customers versus share of identified-customer gross value (%). Source: UCI Online Retail dataset; author calculations.

- Champions are 23.5% of customers but 64.4% of value; Lapsed customers are 22.3% of customers but 4.7% of value. **[Direct]**
- 212 frequent buyers have gone quiet (average 129 days since last order), worth £393,145 so far. Whether a win-back succeeds is **[No evidence]**.
- Segments use recency and frequency scored 1 to 5; customers with identical values always receive the same score. Thresholds are analyst-defined, not a validated model.

Segment	Customers	% of customers	Gross value	% of value	Avg days since last order	Avg invoices
Champions (recent, frequent)	1,018	23.5%	£5,743,280	64.4%	12	10.8
Core (middle)	1,910	44.0%	£2,277,223	25.6%	75	2.7
Lapsed, low frequency	966	22.3%	£422,502	4.7%	222	1.0
At risk (frequent, lapsing)	212	4.9%	£393,145	4.4%	129	5.5
Recent, low frequency	232	5.3%	£75,258	0.8%	19	1.0

Segment rules: Champions = recency and frequency scores 4 or 5. At risk = recency 1 or 2 with frequency 4 or 5. Recent, low frequency = recency 4 or 5 with frequency 1 or 2. Lapsed = both 1 or 2. Core = the rest. Method follows Chen, Sain and Guo (2012), the dataset's introductory paper.

4. Cancellations, value leakage and data risk

Question: How much value is lost to cancellations, and where does the risk sit?
Answer: 8.4% of gross value (£896,812) sits on 3,836 cancellation invoices (9,288 rows). The exposure is concentrated: the top 1% of cancellation invoices (38) hold 64.9% of it. Order value alone predicts cancellation weakly; single lines above 5,000 units were cancelled in 2 of 2 cases. **[Direct]**

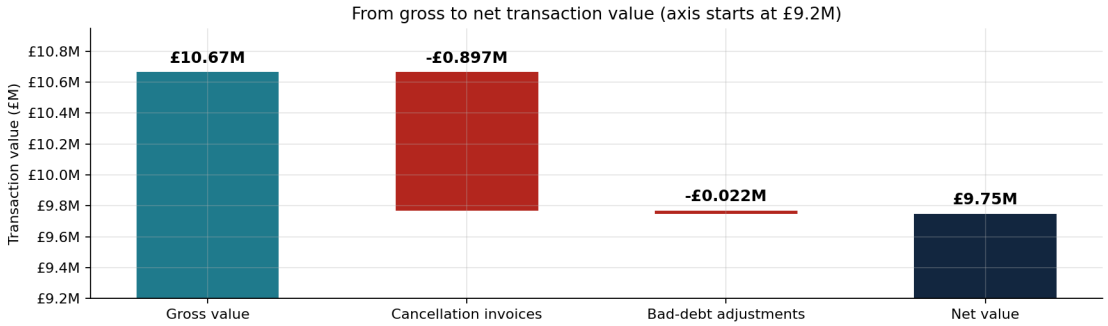


Figure 5. Bridge from gross to net transaction value (£M; axis starts at £9.2M). Source: UCI Online Retail dataset; author calculations.

- Gross £10,666,685 less cancellation invoices £896,812 (8.4%) and two bad-debt adjustments £22,124 gives net £9,747,748. **[Direct]**
- The bridge explains the whole gap (unexplained difference £0.00).
- The two adjustment lines are described "Adjust bad debt" at -£11,062.06 each.

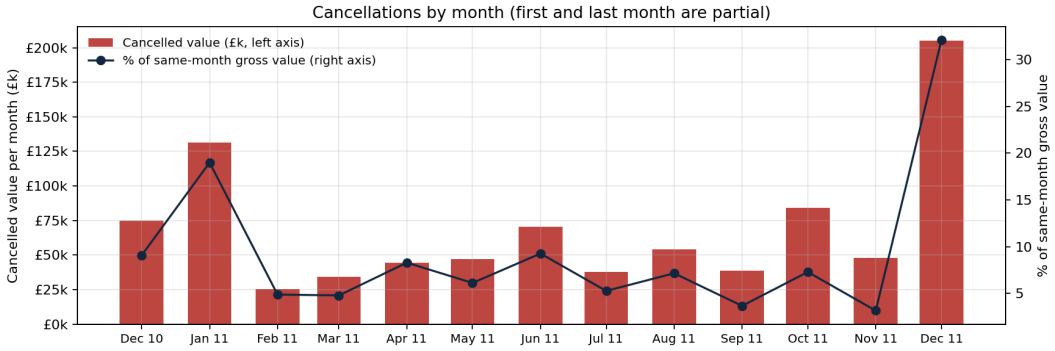


Figure 6. Cancelled value by month (£ thousand) and as a percentage of same-month gross value. Source: UCI Online Retail dataset; author calculations.

- Cancellations run at 3.2% to 19.0% of same-month gross value in full months; the first and last months are partial.
- Cancellations are recorded in the month they occur, so the ratio mixes months when orders were placed earlier; read it as an indicator, not a rate.
- The two largest cancelled orders fall in Jan 2011 and Dec 2011, the two months with the highest cancelled value. **[Direct]**

Order size and cancellation risk

Invoice values (gross, 19,960 invoices) are skewed: the median is £303.84, so it cannot serve as a "large order" threshold (half of all invoices exceed it). Percentiles give a defensible threshold.

Percentile	Invoice value	Invoices above	Share of gross value above
P50	£303.84	9,980	85.7%
P75	£495.61	4,990	68.1%
P90	£940.89	1,996	49.4%
P95	£1,585.92	998	38.2%
P99	£4,821.17	200	19.3%
P99.9	£16,495.76	20	6.5%
Max	£168,469.60	-	-

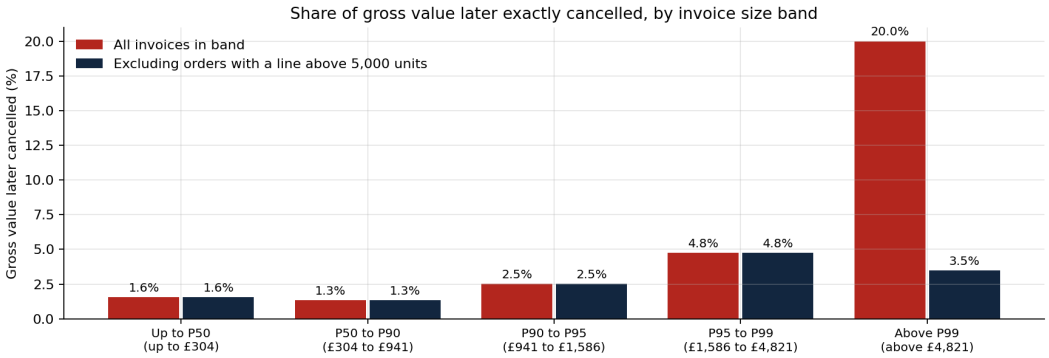


Figure 7. Share of gross value later exactly cancelled, by invoice size band (%). Source: UCI Online Retail dataset; author calculations.

- Up to P99 the cancelled share rises only modestly: 1.6%, 1.3%, 2.5%, 4.8%. **[Direct]**
- Above P99 it is 20.0%, but 3.5% once the 2 orders with a line above 5,000 units are excluded: order value alone is a weak predictor.
- A P99 review (£4,821) flags 200 invoices, about 4 a week, covering 19.3% of gross value. It is justified by exposure, not by higher cancellation odds.

The only 2 gross lines above 5,000 units

Order	Cancelled by	Product	Units	Value	Placed	Minutes to cancel
581483	C581484	23843	80,995	£168,469.60	09 Dec 2011	12
541431	C541433	23166	74,215	£77,183.60	18 Jan 2011	16

Both were cancelled by the same customer within 12 to 16 minutes, for £245,653 (2.3% of gross value). With only 2 cases, whether such a rule generalises is **[Hypothesis]**. 2 further lines above 5,000 units have a zero price and are outside the gross measure.

At least 3,887 gross lines worth £445,875 (4.2% of gross value) have a later cancellation with the same customer, product, quantity and price. This is a lower bound (partial returns and lines without a CustomerID are not captured). **[Direct]** The reason for cancellation is **[No evidence]**.

Data-quality indicators

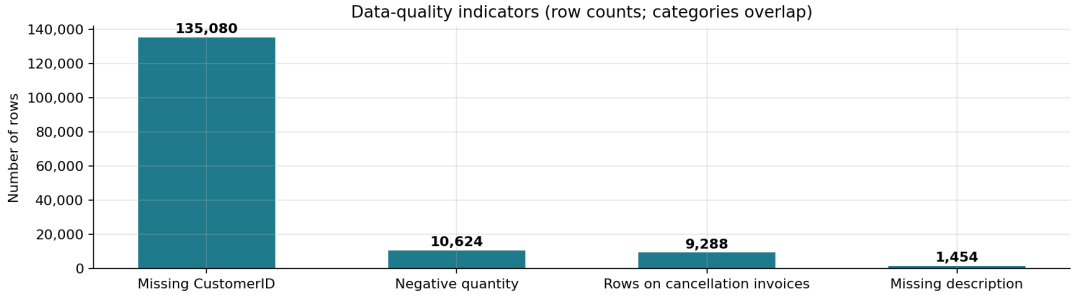


Figure 8. Row counts per data issue (categories overlap, so bars must not be added). Source: UCI Online Retail dataset; author calculations.

- 135,080 rows (24.9%) have no CustomerID; 10,624 have negative quantity; 9,288 sit on cancellation invoices; 1,454 lack a description.
- The UCI documentation says the dataset has no missing values, which the data contradicts. Two invoice numbers start with "A", a prefix the documentation does not explain. **[Direct]**

Indicator	Count	Why it matters
Zero unit price rows	2,515	Samples, promotions or adjustments; unverified
Negative unit price rows	2	Both are bad-debt adjustments
Missing invoice date	0	None

5. Recommendations and action plan

Question: What should we do first?
Answer: Fix the reporting basis (R1), control extreme-quantity orders (R2), then prepare for the autumn peak (R3) and protect core customers (R4). Each recommendation below carries its supporting KPI and evidence level.

Sizes show how much is at stake in this dataset. Illustrative scenarios are arithmetic, not forecasts. Priority reflects value at stake and effort (analyst judgement); owners and efforts are suggestions.

#	Recommendation and first step	Supporting KPI / finding	Evidence	Value at stake and scenario	Owner, effort
R1	**Report net value beside gross value** and publish one set of definitions. First step: adopt Appendix A in every report.	Gross is £918,937 (9.4%) above net; product 23843 has £168,470 gross, £0 net.	[Direct]	£918,937 of overstatement avoided in every report.	Finance, BI Low
R2	**Confirm any single line above 5,000 units before dispatch**, and review invoices above P99 (£4,821) monthly. First step: set the rule and log each confirmation.	2 lines above 5,000 units, both cancelled within 12 to 16 minutes (£245,653). Top 1% of cancellation invoices = 64.9% of cancelled value. P99 review = 200 invoices a year.	[Direct] facts [Hypothesis] rule reduces cancellations	£896,812 cancelled. Each 10% cut in cancelled value = £89,681 (scenario).	Operations, Sales Low to medium
R3	**Plan stock, staff and cash for September to November.** First step: build the plan from August; collect 2012 data to confirm.	Sep to Nov = 34.9% of gross value; November = 2.2x Jan to Aug average; September +39.4%.	[Direct] 2011 pattern [No evidence] repeats	£3.72M of gross value falls in the peak quarter.	Supply chain, Finance Medium
R4	**Protect the core customer group.** First step: key-account cover for the ten largest customers; contact the at-risk list and track response.	Champions: 23.5% of customers = 64.4% of value. Top ten = 17.3%. At risk: 212 customers, £393,145.	[Direct] size [No evidence] win-back works	£393,145 at-risk. Each 10% recovered = £39,314 (scenario).	Sales, CRM Medium
R5	**Close the customer-visibility gap.** First step: find out why IDs are missing (checkout flow, phone or wholesale orders) before changing anything.	24.9% of lines and 16.5% of gross value (£1,755,277) have no CustomerID.	[Direct] size [No evidence] cause	£1,755,277 cannot be tied to buyers.	IT, E-commerce Medium
R6	**Test whether international value can grow.** First step: compare growth and basket size by market using more than one year of data.	Non-UK markets = 15.4% of gross value (£1,641,462); top three: Netherlands, EIRE, Germany.	[Direct] share [No evidence] growth	£1,641,462 is the current non-UK base.	Commercial Medium to high

Suggested 30, 60 and 90 day sequence

Window	Actions	KPI to watch
Days 1 to 30	R1 definitions and net reporting; R2 rule for lines above 5,000 units; start the CustomerID cause review (R5).	Gross-to-net gap (8.6% of gross); lines above 5,000 units confirmed before dispatch.
Days 31 to 60	R3 peak-season plan from August data; R4 key-account cover and at-risk contact list.	Cancelled value (8.4% of gross); at-risk customers contacted (212).
Days 61 to 90	R5 fix for ID capture; R6 market comparison design; review R2 results.	Gross value with a CustomerID (83.5%); non-UK share (15.4%).

Baseline KPIs from this data

KPI	Baseline	Source
Net transaction value	£9.75M	Bridge, Figure 5
Cancelled value as share of gross	8.4%	Section 4
Share of gross value with a CustomerID	83.5%	Section 3
Repeat customers	65.6%	Section 3
Champions share of identified gross value	64.4%	Section 3
September to November share of gross value	34.9%	Section 1

6. Assumptions, limitations and data handling

Assumptions

- Calculated transaction value (Quantity x UnitPrice) is not audited revenue, profit or cash collected. No tax, shipping, discount or accounting treatment is applied.
- An InvoiceNo starting with "C" is a cancellation invoice, as stated in the dataset documentation. **[Direct]**
- Each row is a transaction line, not an order or a customer. The same customer can appear on many lines and invoices.
- DOT, M and POST are treated as service codes in the product view only. This rule needs business confirmation.
- Analyst-defined RFM thresholds describe the data; they are not a validated predictive model.

How cancellations and duplicates are handled

Topic	Treatment in this report	Effect
Cancellation invoices	Excluded from gross value; included in net value. 3,836 invoices, 9,288 rows, £896,812.	Net is 8.6% below gross.
Matching cancellations to orders	A gross line counts as reversed only if a later cancellation line has the same customer, product, quantity and price (identified customers).	Lower bound: at least 4.2% of gross value.
Partial returns, lines without CustomerID	Not matched to orders.	Understates reversals.
Negative quantity on non-cancellation invoices	1,336 rows; all have zero price, so they add no value.	None on value.
Exact duplicate rows	5,268 rows (0.97% of rows) are exact repeats of an earlier row. They are retained because the data has no line identifier to show whether they are errors or repeated scans.	Removing them would change gross by -0.23% (£24,574) and net by -0.22%: immaterial.
Rows without CustomerID	Kept in all value totals; excluded from customer-level analysis.	£1,755,277 cannot be attributed.

Limitations

- **One year of data.** Seasonality, trend and customer behaviour cannot be separated or confirmed as recurring.
- **No cost, margin or customer-type data.** Findings describe transaction value, not profit.
- **Relevance today.** The data covers December 2010 to December 2011. Levels in pounds, markets and customer behaviour may differ now; the method, the data-quality lessons and the shape of the risks (concentrated cancellations, concentrated customers, unidentified customers) are more likely to transfer. This is analyst judgement, not evidence from the data.
- Country and monthly views include service codes; the product view excludes them, so those views are not perfectly comparable.
- The invoice-size analysis covers identified customers only (lines with a CustomerID); percentile thresholds use all gross invoices.
- Only descriptive analysis was performed. No causal claims are made.

Appendix A. Definitions and dataset

Exact definitions

Term	Definition	Exact rule (SQL)
Gross transaction value	Calculated value of positive-quantity, positive-price lines that are not on cancellation invoices. Called "positive-item value" in the analysis scripts.	SUM(Quantity * UnitPrice) WHERE InvoiceNo NOT LIKE 'C%' AND Quantity > 0 AND UnitPrice > 0
Net transaction value	Calculated value of every row: gross value less cancellations and adjustments. Called "all-rows net calculated value" in the scripts.	SUM(Quantity * UnitPrice) over all rows
Cancellation value	Value of rows on cancellation invoices, shown as a positive number.	-SUM(Quantity * UnitPrice) WHERE InvoiceNo LIKE 'C%'
Cancellation invoice / row	An invoice number starting with C, and a line on such an invoice. One invoice has several rows.	COUNT(DISTINCT InvoiceNo) versus COUNT(*) WHERE InvoiceNo LIKE 'C%'
Reversed line	A gross line with a later cancellation line for the same customer, product, quantity and price.	EXISTS (cancellation line: same CustomerID, StockCode, UnitPrice, Quantity = -Quantity, later date)
Invoice value	Gross value summed per invoice. Percentiles use all gross invoices.	SUM(Quantity * UnitPrice) GROUP BY InvoiceNo, gross filter
Identified customer	A row with a non-missing CustomerID.	CustomerID IS NOT NULL
Customer who bought	An identified customer with at least one gross line.	DISTINCT CustomerID with gross lines
Top 20% by value	Customers ranked by total gross value; the highest 20% by count.	ROW_NUMBER() OVER (ORDER BY value DESC) <= 20% of customers
Champions	Recency and frequency scores both 4 or 5 (see Section 3).	Average-rank scoring in 5 bands

Dataset

UCI Machine Learning Repository, Online Retail: all transactions of a UK-based, UK-registered non-store online retailer between 1 Dec 2010 and 9 Dec 2011; 541,909 rows and 8 fields. The company mainly sells all-occasion gifts and many customers are wholesalers. UnitPrice is in sterling and an InvoiceNo starting with "c" marks a cancellation, per the documentation.

Field	Role / interpretation
InvoiceNo	Invoice identifier; "C" prefix marks a cancellation.
StockCode	Product or item code; DOT, M and POST behave like service items.
Description	Item name; missing on some rows; some codes have several descriptions.
Quantity	Units on the line; negative for returns, cancellations or adjustments.
InvoiceDate	Transaction timestamp used for monthly analysis.
UnitPrice	Price per unit in sterling; zero and negative values occur.
CustomerID	Customer identifier; missing on about a quarter of rows.
Country	Country where the customer resides.

References

Chen, D. (2015). Online Retail [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5BW33 (CC BY 4.0).

Chen, D., Sain, S. L., & Guo, K. (2012). Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining. Journal of Database Marketing & Customer Strategy Management, 19(3).

Appendix B. Verification

Method. Every KPI in this report was computed twice by independent code paths, DuckDB SQL and pandas, and the results were compared (tolerance 0.005 for pounds and percentages, exact for counts). **48 of 48 KPIs match.** Customer segments were also computed twice: SQL window functions (average rank) against pandas ranks. The report text itself was then checked against the raw data by a third script that recomputed each figure and searched the PDF for it.

Data used. Verification was run on a public copy of the UCI dataset (R package "onlineretail"). Its headline figures equal those from the original workbook run (541,909 rows; 8 columns; 4,372 customers; 135,080 lines without CustomerID; gross value 10,666,684.54; top product 22423), which is indirect evidence that the copy equals the original **[Indirect]**. Re-running verify_analysis.py on the original workbook is the final confirmation step.

#	KPI	DuckDB SQL	pandas	Result
1	Rows	541,909	541,909	Match
2	Columns	8	8	Match
3	Distinct invoice IDs	25,900	25,900	Match
4	Distinct identified customers	4,372	4,372	Match
5	Distinct stock codes	4,070	4,070	Match
6	Rows missing CustomerID	135,080	135,080	Match
7	Rows missing CustomerID (%)	24.93%	24.93%	Match
8	Rows missing Description	1,454	1,454	Match
9	Rows missing InvoiceDate	0.00	0.00	Match
10	Rows with negative quantity	10,624	10,624	Match
11	Rows with zero unit price	2,515	2,515	Match
12	Rows with negative unit price	2	2	Match
13	Exact duplicate rows (beyond the first)	5,268	5,268	Match
14	Cancellation rows (InvoiceNo starts with C)	9,288	9,288	Match
15	Cancellation invoices (distinct)	3,836	3,836	Match
16	Cancellation value (GBP)	896,812.49	896,812.49	Match
17	Top 1% of cancellation invoices: share of cancelled value (%)	64.87%	64.87%	Match
18	Gross lines with an exact later cancellation (count)	3,887	3,887	Match
19	Gross lines with an exact later cancellation (GBP)	445,874.74	445,874.74	Match
20	Gross lines above 5,000 units	2	2	Match
21	Gross transaction value	10,666,684.54	10,666,684.54	Match
22	Net transaction value	9,747,747.93	9,747,747.93	Match
23	Gross minus net	918,936.61	918,936.61	Match
24	Gross minus net, as % of gross	8.62%	8.62%	Match
25	Gross above net, as % of net	9.43%	9.43%	Match
26	United Kingdom gross value	9,025,222.08	9,025,222.08	Match
27	United Kingdom share of gross (%)	84.61%	84.61%	Match
28	United Kingdom net value	8,187,806.36	8,187,806.36	Match
29	November 2011 gross value	1,509,496.33	1,509,496.33	Match
30	January to August 2011 average monthly gross	685,134.99	685,134.99	Match
31	September to November share of gross (%)	34.90%	34.90%	Match
32	Invoices with gross lines	19,960	19,960	Match
33	Average invoice value	534.40	534.40	Match
34	Median invoice value (P50)	303.83	303.83	Match
35	P95 invoice value	1,585.92	1,585.92	Match
36	P99 invoice value	4,821.17	4,821.17	Match
37	Identified customers with gross lines	4,338	4,338	Match
38	Identified customers without gross lines	34	34	Match
39	Gross value on rows without CustomerID	1,755,276.64	1,755,276.64	Match
40	Top 20% of customers: share of identified gross value (%)	74.62%	74.62%	Match
41	Top 10 customers: share of identified gross value (%)	17.26%	17.26%	Match
42	Repeat customers, 2 or more invoices (%)	65.58%	65.58%	Match
43	Champions: customers	1,018	1,018	Match
44	Core: customers	1,910	1,910	Match
45	Lapsed: customers	966	966	Match
46	At risk: customers	212	212	Match
47	Recent, low frequency: customers	232	232	Match
48	Champions: share of identified gross value (%)	64.45%	64.45%	Match

Appendix C. Technical scope and project files

Tools: Python, Pandas, DuckDB SQL, Matplotlib, Google Colab, ReportLab.

Business question	Approach demonstrated
What does the dataset contain?	Row and column counts, distinct values, missing-value and duplicate checks.
How does value vary by month?	DATE_TRUNC monthly aggregation, LAG for change, running totals.
Which countries and products lead?	Grouped aggregation, join to a de-duplicated description lookup, ranking with window functions.
Where is value lost?	Cancellation matching with EXISTS and self-join, percentile bands, concentration shares, gross-to-net bridge.
Who are the valuable customers?	Customer-level aggregation, concentration shares, RFM scoring with window functions.
Can the numbers be trusted?	Every KPI computed twice (SQL and pandas), sensitivity to duplicates, text-to-data audit.

SQL skills demonstrated

Skill area	Constructs used
Basic querying	SELECT, WHERE, DISTINCT, ORDER BY, LIMIT
Aggregation	COUNT, COUNT(DISTINCT), SUM, AVG, MIN, MAX, MEDIAN, quantile_cont, FILTER clause
Grouping and conditions	GROUP BY, HAVING, CASE WHEN, COALESCE
Date analysis	DATE_TRUNC, EXTRACT, DATE_DIFF, epoch arithmetic
Joins	LEFT JOIN ... USING, self-join for cancellation matching, de-duplicated lookup (MODE per StockCode)
CTEs and subqueries	Multi-step CTEs, scalar and IN subqueries, correlated EXISTS
Window functions	ROW_NUMBER, RANK, DENSE_RANK, LAG, LEAD, COUNT OVER, running totals with frame clause

Project files

Deliverable	Contents
Analysis notebook (Google Colab)	Reproducible SQL and Python workflow, from loading the workbook to exports.
final_analysis.py	All CSV tables and the eight report charts.
verify_analysis.py	Independent SQL-versus-pandas verification of every KPI (writes verification_results.csv).
build_final_report.py	Builds this PDF from the exported tables; fails if a table or chart is missing.
README	Objective, tools, metric definitions, limitations and findings.

Portfolio description

Analysed 541,909 online-retail transaction lines (public UCI dataset) with Python, Pandas, DuckDB SQL and Matplotlib. Built a gross-to-net value bridge, monthly and seasonal analysis, customer concentration and RFM segmentation, and a cancellation analysis that matched cancelled orders to their originals (finding that 8.4% of gross value was cancelled, concentrated in a few invoices). Verified every KPI twice with independent SQL and pandas code, documented cancellation and duplicate handling, and turned the results into a ranked action plan with evidence levels.

Verification & Corrections Addendum

Supplement to Online Retail Transaction Analysis | Executive report | Prepared 5 November 2026

This addendum is attached to the original 12-page report. The original report pages are preserved unchanged; the notes below clarify the verification notebook results and supersede the affected technical explanations.

1. Dataset structure check

The source workbook contains **541,909 rows and 8 original columns**. The verification notebook adds two derived helper columns during processing: *line_value* and *is_cancel*. Therefore, a later len(df.columns) check returns 10. This is not a source-data problem. The correct source-column check must be captured immediately after reading the Excel file, before adding helper columns.

Check	Expected	Notebook output	Interpretation
Original workbook columns	8	8	Correct source structure
Columns after 2 helper fields	—	10	Expected processed DataFrame width
Other data-quality checks	13 checks	13 matched	All remaining checks shown matched

2. Correct gross-to-net reconciliation

The first verification run exposed a formula issue in the addendum notebook—not in the original workbook. The formula counted the negative *Adjust Bad Debt* lines twice. The corrected bridge contribution is **£22,124.12**, not £33,186.18. The corrected reconciliation is:

Bridge component	Amount
Gross positive-item transaction value	£10,666,684.54
Less: cancellation invoice value (absolute)	(£896,812.49)
Less: Adjust Bad Debt bridge contribution	(£22,124.12)
Net value across all rows	**£9,747,747.93**

The £22,124.12 is the bridge contribution needed to reconcile the gross-positive-item measure to the all-row net measure. It is **not** the net bad-debt expense. The three Adjust Bad Debt records have a net signed impact of –£11,062.06; the bridge contribution differs because the gross measure includes the positive Adjust Bad Debt row while the all-row net measure includes all three records.

Corrected notebook expression: gross-value contribution from Adjust Bad Debt rows minus the signed total of all Adjust Bad Debt rows. Re-run the financial-check cell after applying this correction; expected bridge difference from the all-row net value is £0.00 (within rounding tolerance).

3. What the submitted verification output establishes

The screenshots supplied from Google Colab support the following conclusions. They do not yet establish that every KPI and every RFM segment in the executive report has been independently validated.

Area	Current evidence	Status
Source shape and data quality	13 of 14 checks matched; the one mismatch is explained by helper columns.	Explained; rerun corrected check
Gross and net transaction values	Gross £10,666,684.54 and net £9,747,747.93 matched reference values.	Matched
Cancellation value	Absolute cancellation value £896,812.49 matched.	Matched
Bad-debt bridge	First run returned £33,186.18 due to notebook formula error.	Correction required and rerun
DuckDB vs Pandas	SQL aggregate output was supplied, but the comparison table with per-metric match flags was not supplied.	Not yet confirmed
RFM segmentation	RFM output table was not supplied; notebook's simple segment rules may differ from the executive report's RFM rules.	Not yet confirmed
All 48 KPIs	The screenshots do not show a complete 48-KPI comparison.	Not yet established

4. Reporting corrections and interpretation

• **StockCode counts:** report 4,070 distinct StockCodes separately from 5,752 distinct StockCode–Description pairs. These are different measures.

• **Invoice P50:** use £303.84 for the interpolated median (continuous 50th percentile); do not mix it with the lower-middle observation £303.83.

• **Large quantities:** retain the recommendation as a controlled review/pilot. Two observations above 5,000 units are insufficient to establish a permanent cancellation rule.

• **RFM:** preserve the executive report's documented segment definitions unless the RFM calculation is rerun using exactly those rules. The verification notebook's illustrative segment rules are not automatically equivalent.

• **Scope of validation:** do not describe all 48 KPIs as independently verified until a full KPI-by-KPI comparison output is available.

5. Final audit status

Core source-data checks and the main gross/net/cancellation amounts are consistent with the independently calculated reference values. The first notebook run also exposed two notebook-only issues: the source column count was measured after adding helper fields, and the bad-debt bridge formula double-counted negative lines. The bridge correction is specified above. SQL-vs-Pandas match flags, report-aligned RFM results, and the complete 48-KPI audit remain unconfirmed by the screenshots supplied so far.

This addendum is a transparent verification note, not a claim of a full external audit. The underlying original 12-page report has been retained without altering its layout.